

Code changes from implicit to CEL

fe-large-displacement

Each folder under `examples/indenter/` is one complete script. Open two of them and diff. This note lists the extra lines at each rung.

1 Implicit

`examples/indenter/implicit/model.py`

Geometrically nonlinear static step

Automatic stabilisation is fictitious viscous damping. Check ALLSD.

```

127 # *Static, geometrically nonlinear, automatic stabilisation (fictitious
128 # viscous damping). Check ALLSD against ALLIE if the job runs far enough.
129 model.StaticStep(
130     name='Push', previous='Initial', nlgeom=ON, timePeriod=1.0,
131     initialInc=0.005, minInc=1.0e-8, maxInc=0.05, maxNumInc=1000,
132     stabilizationMethod=DISSIPATED_ENERGY_FRACTION,
133     stabilizationMagnitude=2.0e-4, adaptiveDampingRatio=0.05,
134     continueDampingFactors=ON)
135 model.TabularAmplitude(name='Ramp', timeSpan=STEP,
136     data=((0.0, 0.0), (1.0, 1.0)))

```

Displacement control

```

165 # Displacement control: push the rigid body down by IND_DEPTH over the step.
166 model.DisplacementBC(
167     name='Push', createStepName='Push', region=asm.sets['IndRP'],
168     u1=0.0, u2=0.0, u3=-IND_DEPTH, ur1=0.0, ur2=0.0, ur3=0.0,
169     amplitude='Ramp', distributionType=UNIFORM)

```

Standard element library

```

103 # First-order reduced-integration hex, Standard library.
104 elem = mesh.ElemType(elemCode=C3D8R, elemLibrary=STANDARD)

```

Standard general contact

```

177 contact = model.ContactStd(name='GeneralContact', createStepName='Push')
178 contact.includedPairs.setValuesInStep(stepName='Push', useAllstar=True)
179 contact.contactPropertyAssignments.appendInStep(
180     stepName='Push', assignments=((GLOBAL, SELF, 'Interface'))))

```

2 Implicit ALE

On top of implicit. `examples/indenter/implicit_ale/model.py`

Restricted adaptive domain

ALE only accepts first-order reduced-integration solids (C3D8R). An empty set is silent. Standard has no initial mesh sweeps.

```

175 # Restrict the domain to a box around the indenter. An empty set is silent:
176 # Abaqus will not warn, and ALE will do nothing.
177
178 half = 3.0 * IND_HALF
179 zmin = max(-1.3 * IND_DEPTH, -SOIL_DEP + 2.0 * SEED)
180 ale_elems = soil_inst.elements.getByBoundingBox(
181     xMin=-half, xMax=half, yMin=-SOIL_WID, yMax=SOIL_WID,
182     zMin=zmin, zMax=0.0)
183 if len(ale_elems) == 0:
184     raise RuntimeError('ALE box is empty. ALE would run and do nothing.')
185 asm.Set(name='ALE_Box', elements=ale_elems)
186 print('ALE elements: %d' % len(asm.sets['ALE_Box'].elements))
187
188 model.AdaptiveMeshControl(name='ALE_BC', smoothingPriority=UNIFORM)
189 # Standard has no initial mesh sweeps. Do not pass initialMeshSweeps.
190 model.steps['Push'].AdaptiveMeshDomain(
191     region=asm.sets['ALE_Box'], controls='ALE_BC',
192     frequency=10, meshSweeps=1)

```

Diagnostics after *Static

CAE cannot write *Diagnostics. Without it you cannot prove ALE ran.

```

204 # CAE cannot write *Diagnostics. This inserts it after *Static.
205 inject_diagnostics(inp, mode='summary', include_static=True)

```

3 Explicit

On top of implicit, replacing the solver. `examples/indenter/explicit/model.py`

Mass (*Density)

Required. The stable increment is $L_{\min}/c_d$.

```

85 soil_mat = model.Material(name='SoilMat')
86 soil_mat.Density(table=((SOIL_RHO,)),)
87 soil_mat.Elastic(table=((SOIL_E, SOIL_NU)),)

```

Explicit element library

```

101 # Same C3D8R, Explicit library.
102 elem = mesh.ElemType(elemCode=C3D8R, elemLibrary=EXPLICIT)

```

Dynamic explicit step, not static

Drop automatic stabilisation. Bulk viscosity is on by default (ALLVD). Do not add extra damping to hide kinetic energy.

```

125 # *Dynamic, Explicit. No automatic stabilisation. Bulk viscosity is on by
126 # default (watch ALLVD). Do not add extra damping to hide kinetic energy.
127 time_period = IND_DEPTH / float(IND_VEL)
128 model.ExplicitDynamicsStep(
129     name='Push', previous='Initial', timePeriod=time_period,
130     nlgeom=ON, improvedDtMethod=ON)
131 model.SmoothStepAmplitude(
132     name='Ramp', timeSpan=STEP,
133     data=((0.0, 0.0), (RAMP_FRAC * time_period, 1.0)))

```

Energy history

Peak ALLKE/ALLIE below about 5 % is acceptable, below 1 % is quotable.

```

134 model.HistoryOutputRequest(
135     name='Energies', createStepName='Push',
136     variables=('ALLIE', 'ALLKE', 'ALLAE', 'ALLDC', 'ALLVD', 'ALLWK',
137               'ETOTAL'),
138     numIntervals=200)

```

Velocity control, not displacement

```

163 # Velocity control, not displacement. Sign is negative: penetration is -z.
164 model.VelocityBC(
165     name='Push', createStepName='Push', region=asm.sets['IndRP'],
166     v1=0.0, v2=0.0, v3=-IND_VEL, vr1=0.0, vr2=0.0, vr3=0.0,
167     amplitude='Ramp', distributionType=UNIFORM)

```

Explicit general contact

```

175 contact = model.ContactExp(name='GeneralContact', createStepName='Push')
176 contact.includedPairs.setValuesInStep(stepName='Push', useAllstar=True)
177 contact.contactPropertyAssignments.appendInStep(
178     stepName='Push', assignments=((GLOBAL, SELF, 'Interface'))))

```

Double precision job

```

186
187 # --- write, check, maybe submit -----
188
189 job = mdb.Job(
190     name=JOB_NAME, model='Model-1', type=ANALYSIS,
191     explicitPrecision=DOUBLE_PLUS_PACK, nodalOutputPrecision=FULL,

```

Optional mass scaling

Commented at the bottom of the Explicit script. Semi-automatic, target Δt . Then re-check ALLKE.

Optional section controls

Also commented. Distortion control or enhanced hourglass, not both. CAE cannot write *Section Controls.

4 Explicit ALE

On top of explicit. `examples/indenter/explicit_ale/model.py`

Adaptive domain with initial sweeps

Same box as implicit ALE. Explicit adds `initialMeshSweeps=5`.

```

175
176 # --- ALE (the only extra block versus explicit/model.py) -----
177
178 half = 3.0 * IND_HALF
179 zmin = max(-1.3 * IND_DEPTH, -SOIL_DEP + 2.0 * SEED)
180 ale_elems = soil_inst.elements.getByBoundingBox(
181     xMin=-half, xMax=half, yMin=-SOIL_WID, yMax=SOIL_WID,

```

```

182     zMin=zmin, zMax=0.0)
183 if len(ale_elems) == 0:
184     raise RuntimeError('ALE box is empty. ALE would run and do nothing.')
185 asm.Set(name='ALE_Box', elements=ale_elems)
186 print('ALE elements: %d' % len(asm.sets['ALE_Box'].elements))
187
188 model.AdaptiveMeshControl(name='ALE_BC', smoothingPriority=UNIFORM)
189 # Explicit has initial mesh sweeps. Default frequency=10, meshSweeps=1.
190 model.steps['Push'].AdaptiveMeshDomain(
191     region=asm.sets['ALE_Box'], controls='ALE_BC',
192     frequency=10, meshSweeps=1, initialMeshSweeps=5)

```

Diagnostics after *Dynamic, Explicit

Then read the .msg. Zero percent of nodes moved means the domain was defined and did nothing.

```

206 # CAE cannot write *Diagnostics. This inserts it after *Dynamic, Explicit.
207 inject_diagnostics(inp, mode='summary')

```

Run one CPU for a study of ALE. Nodes shared between parallel domains are nonadaptive.

5 Explicit CEL

On top of explicit, replacing the Lagrangian soil. `examples/indenter/explicit_cel/model.py`

Eulerian part with void above the soil

Do not set the void to zero. Heave needs empty elements.

```

70 # Eulerian box: soil depth PLUS empty space above it for heave.
71 H = VOID
72 sketch = model.ConstrainedSketch(name='eul', sheetSize=10.0 * SOIL_LEN)
73 sketch.rectangle(point1=(-SOIL_LEN / 2.0, -SOIL_WID / 2.0),
74                 point2=(SOIL_LEN / 2.0, SOIL_WID / 2.0))
75 eul = model.Part(name='Eulerian', dimensionality=THREE_D, type=EULERIAN)
76 eul.BaseSolidExtrude(sketch=sketch, depth=SOIL_DEP + H)
77
78 # Partition at the soil surface so the filled region is a cell, not a guess.
79 if H > 0.0:
80     datum = eul.DatumPlaneByPrincipalPlane(principalPlane=XYPLANE,
81                                             offset=SOIL_DEP)
82     eul.PartitionCellByDatumPlane(datumPlane=eul.datums[datum.id],
83                                  cells=eul.cells)

```

Eulerian section and EC3D8R

```

99 # Eulerian section, not a solid section. The instance name is the EVF handle.
100 eul_key = 'soilmat-1'
101 model.EulerianSection(name='EulSec', data={eul_key: 'SoilMat'})
102 eul.SectionAssignment(region=regionToolset.Region(cells=eul.cells),
103                     sectionName='EulSec')
104 model.HomogeneousSolidSection(name='IndSec', material='SteelMat')
105 indenter.SectionAssignment(region=regionToolset.Region(cells=indenter.cells),
106                          sectionName='IndSec', offsetType=MIDDLE_SURFACE,
107                          thicknessAssignment=FROM_SECTION)
108
109 eul.setElementType(
110     regions=(eul.cells,),
111     elemTypes=(mesh.ElemType(elemCode=EC3D8R, elemLibrary=EXPLICIT),))

```

Initial volume fraction

Cells below $z = 0$ start full. This is `*Initial Conditions, type=VOLUME FRACTION`.

```

131 # Cells below z = 0 start full. Everything else starts empty.
132 filled = eul_inst.cells.getByBoundingBox(
133     zMin=-SOIL_DEP - 1e-6, zMax=0.0 + 1e-6)
134 if len(filled) == 0:
135     raise RuntimeError('no Eulerian cells below z = 0')
136 filled_set = asm.Set(name='SoilFilled', cells=filled)
137 model.MaterialAssignment(
138     name='Eulerian material assignment',
139     instanceList=(eul_inst,),
140     useFields=False,
141     assignmentList=((filled_set, (1.0,)),))

```

Walls, free top

Zero normal motion on the sides and base. The top is open.

```

173 # Walls: zero normal motion on sides and base. The top is open.
174 top = H
175 for name, point, dof in (
176     ('XMIN', (-SOIL_LEN / 2.0, 0.0, (top - SOIL_DEP) / 2.0), 1),

```

EVF on the field output

```

169 model.fieldOutputRequests['F-Output-1'].setValues(
170     variables=('S', 'PE', 'LE', 'U', 'V', 'RF', 'EVF', 'STATUS'),
171     numIntervals=20)

```

No geostatic, no ALE

`*Geostatic` has no Eulerian implementation. General contact is mandatory. There is no `*Adaptive Mesh` block.

6 What to check in the deck

Rung	The deck must show
Implicit	<code>*Static</code> , <code>nlgeom=YES</code> , displacement BC
Implicit ALE	plus <code>*Adaptive Mesh</code> on <code>ALE_Box</code>
Explicit	<code>*Density</code> , <code>*Dynamic</code> , <code>Explicit</code> , <code>type=VELOCITY</code> , <code>ALLKE</code>
Explicit ALE	plus <code>*Adaptive Mesh</code> , <code>*Diagnostics</code>
Explicit CEL	<code>*Eulerian Section</code> , <code>EC3D8R</code> , volume fraction, void, no ALE

Write the input. Read it. Then submit.